

Validation of an accessible tool for component-based multidisciplinary analysis of electric fixed-wing UAVs



Sammy Nassau

MEng

Kenny Xu

Undergrad

Dr. Jason Hicken

Professor



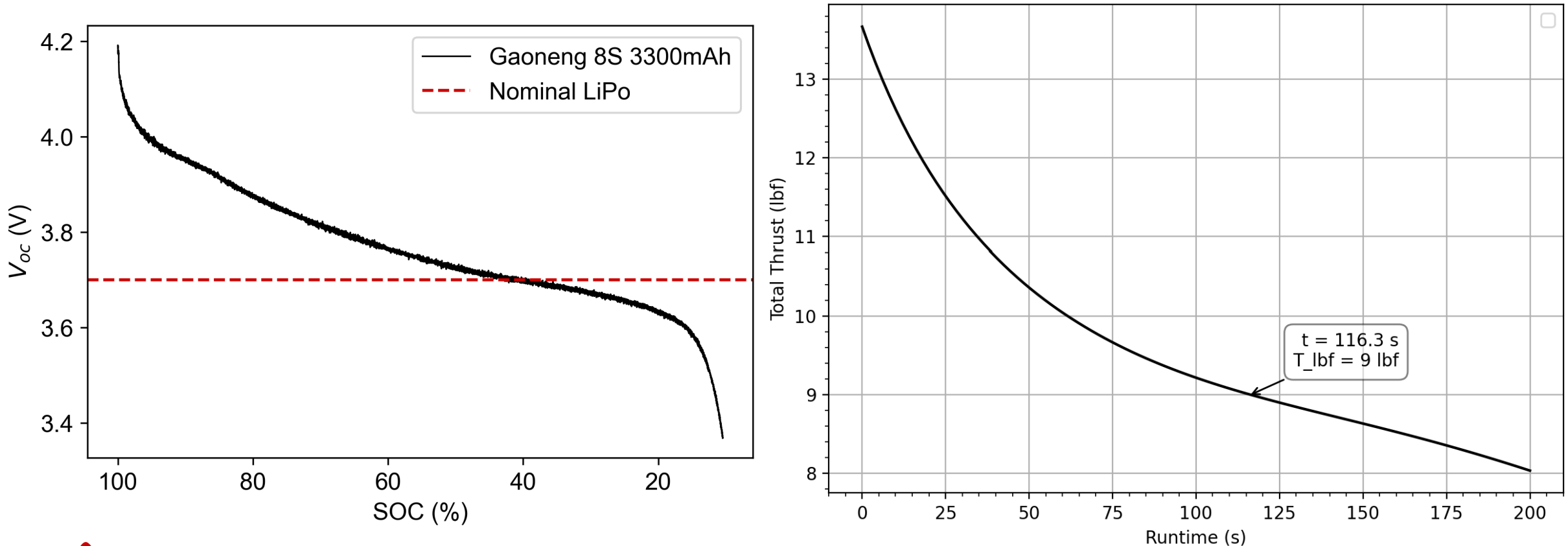
Tucson, AZ. April 2025

29th DBF Fly-Off



“Batteries Are Not Fuel”

- Electric drives produce significantly less thrust over longer runtimes
- Decrease in sustained speed near the end of the mission
- Using the nominal cell voltage is insufficient for propulsion selection



Project Goals

Develop an accessible tool for low-fidelity predictions of electric drive performance over the flight envelope set by U_∞ , δT , h , and t

Subtasks:

- a. Create a propulsion model with straightforward implementation in multidisciplinary design optimization and analysis (MDAO) codes
- b. Validate propulsion model ($\leq 15\%$ error desired)
- c. Create an easy-to-use open-source software

Propulsion Modeling

Propeller Data
(APC BEMT or UIUC exp)

3-Constant Motor Model
 (K_V, I_0, R_m)

Constant ESC Efficiency
 (η_c)

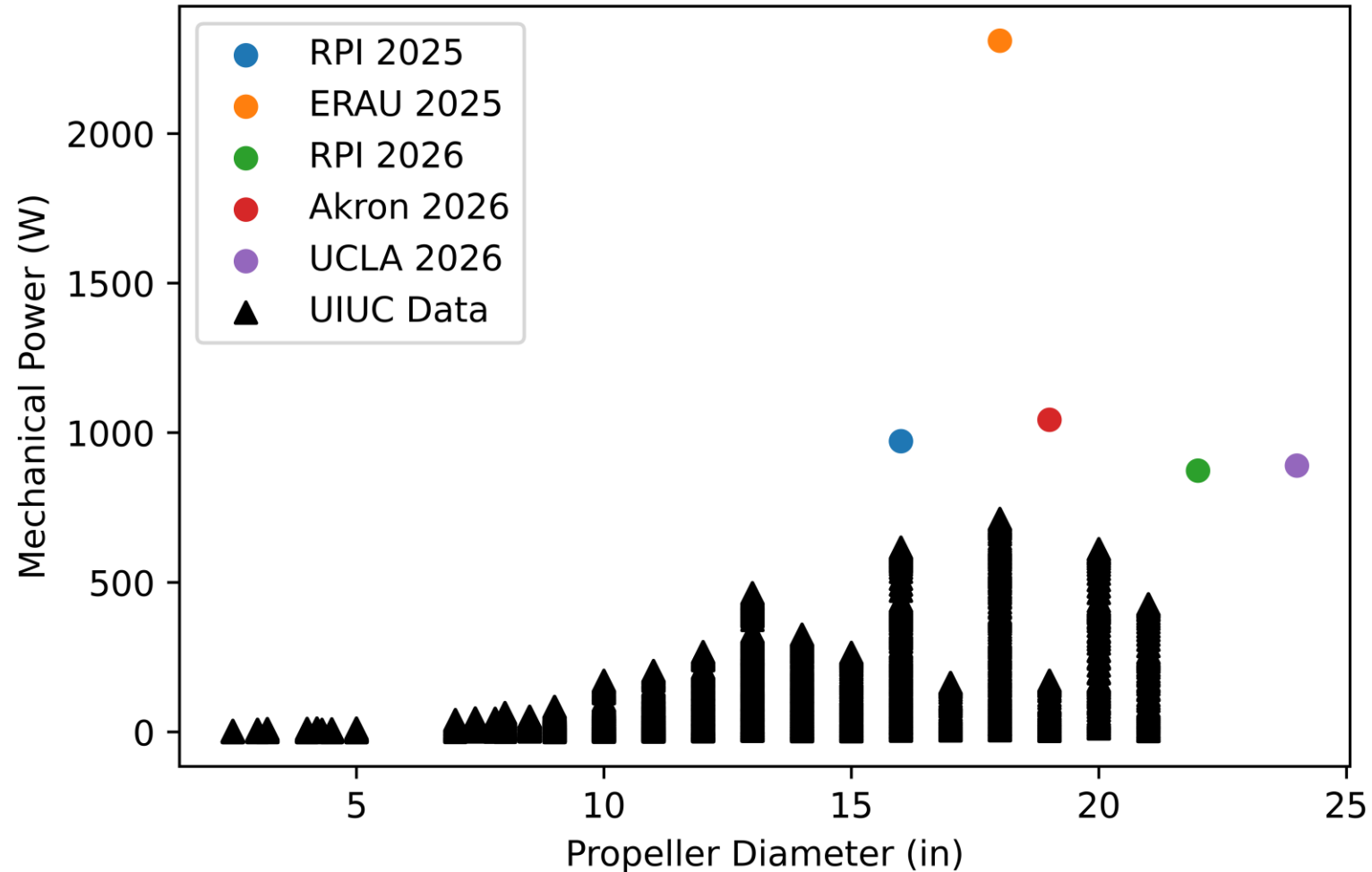
Battery Model
 $(V_{oc} = f(\text{SOC}))$



SimpleRPM Model

Limitations of UIUC Data

APC BEMT is required for setups with >800 W of mechanical power



SimpleRPM

$$\text{Res}(\text{RPM}_p) = \text{RPM}_p - \widehat{\text{RPM}}_p(\text{RPM}_p)$$

where $\widehat{\text{RPM}}_p$ depends on constants defined from component selection and the four flight condition variables; $U_\infty, \delta T, h, t$

Full derivation in the paper!

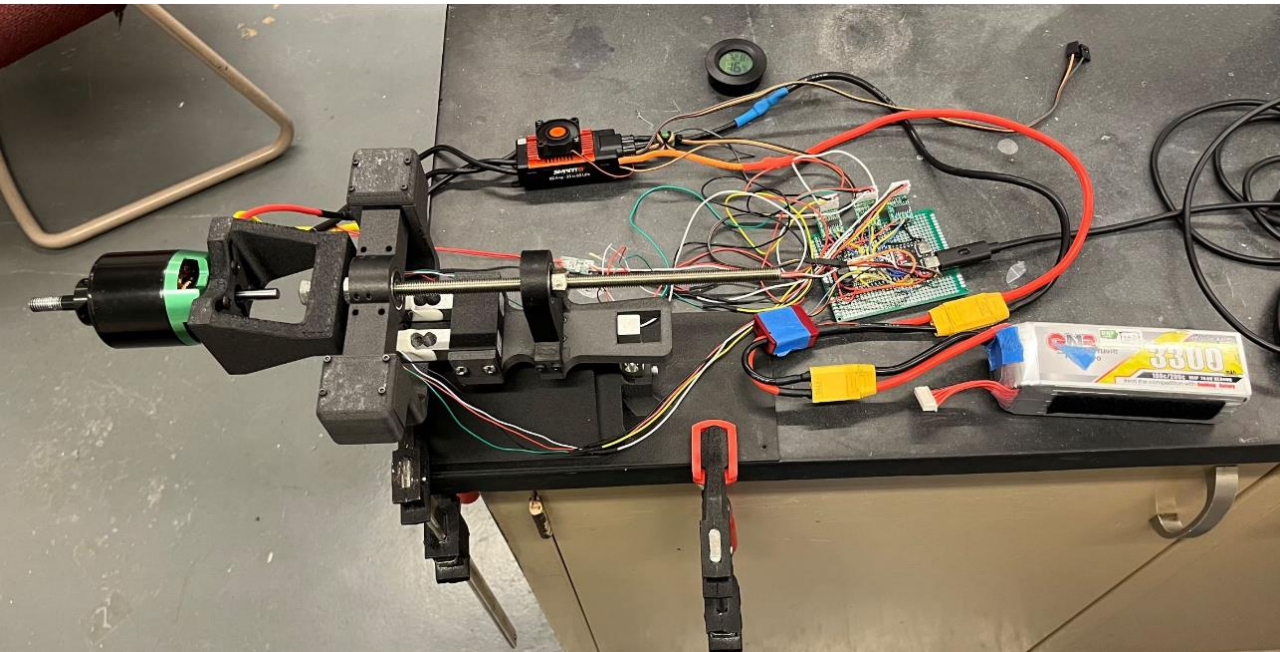
Known RPM_p plus propeller data enables immediate calculation of thrust, torque, efficiencies, and electrical quantities

Validation Efforts

Static testing: 7 propellers, 2 motors, 24 runs

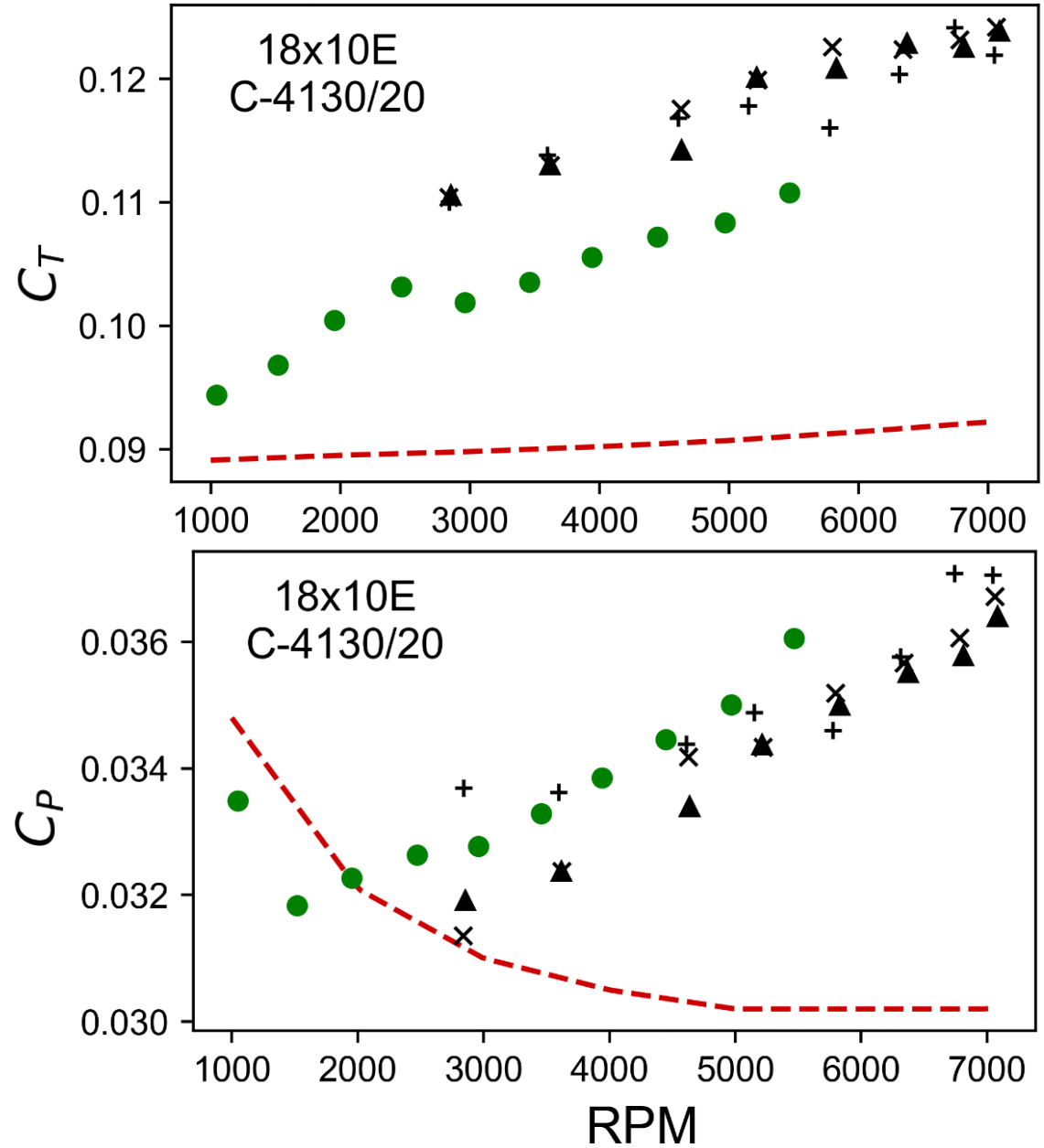
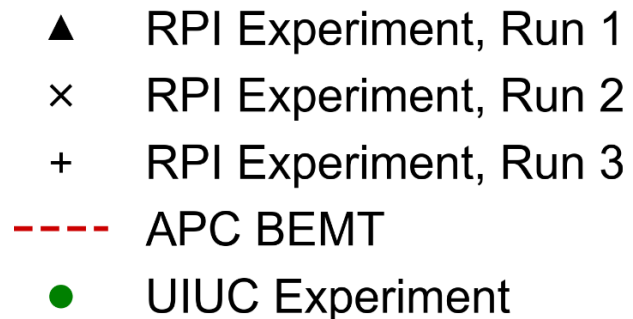
Dynamic testing: 1 run

Flight testing: 3 flights



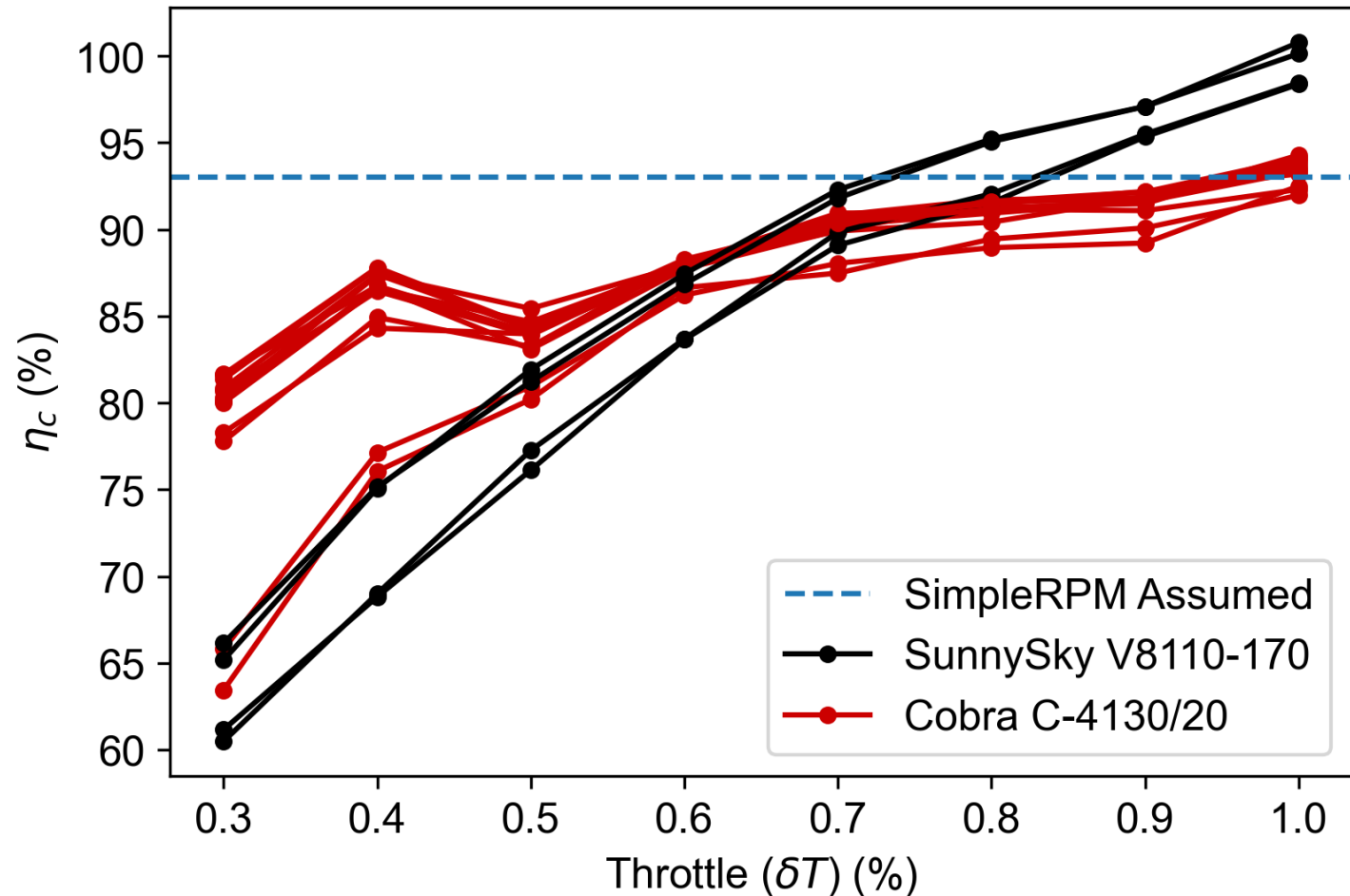
Propeller Data Error

- Representative C_T , C_P results for static condition
- RPI experiments represented satisfactory agreement (<8%) with UIUC data
- APC BEMT significantly underpredicts C_T and C_P , especially at higher RPMs

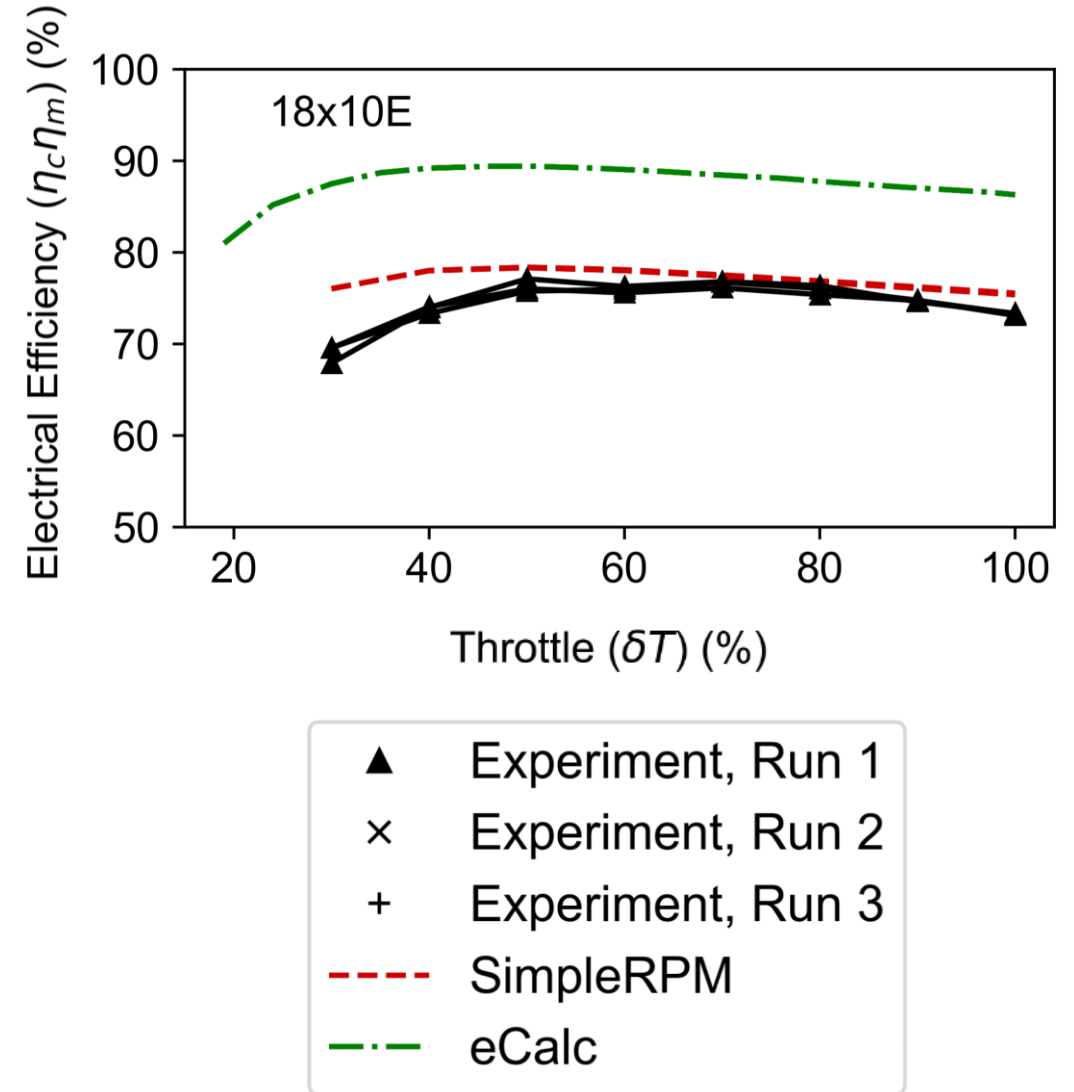
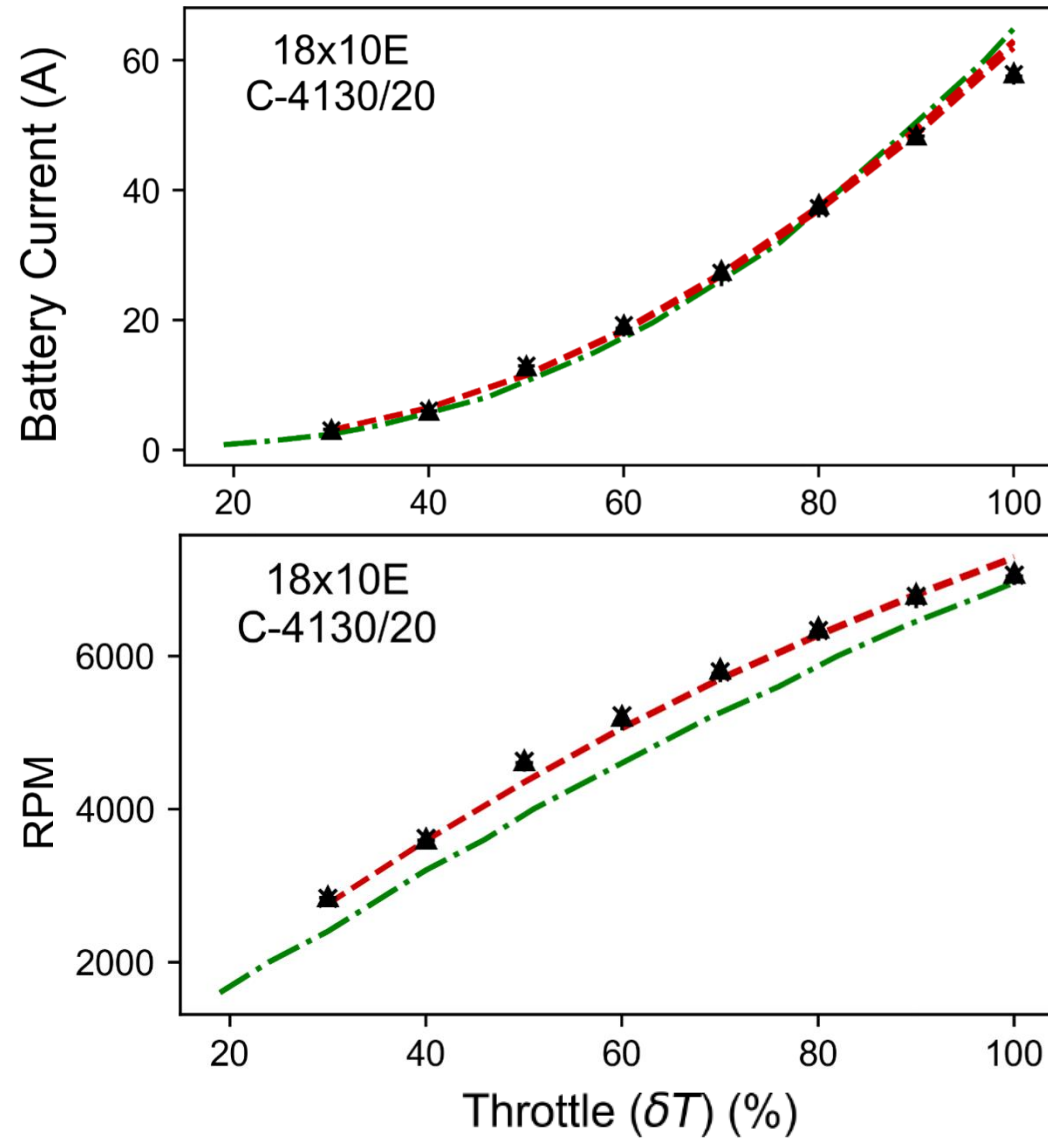


Constant I_0 , η_c and Battery Model Errors

- I_0 error resulted in I_b error $< 1A$ for all tested setups
- $\eta_c = 93\%$ accurate to $< 10\%$ for a Spektrum Avian 80A ESC at $\delta T > 60\%$
- Jeong et al. LiPo battery model had 1% median RE and 4% maximum RE



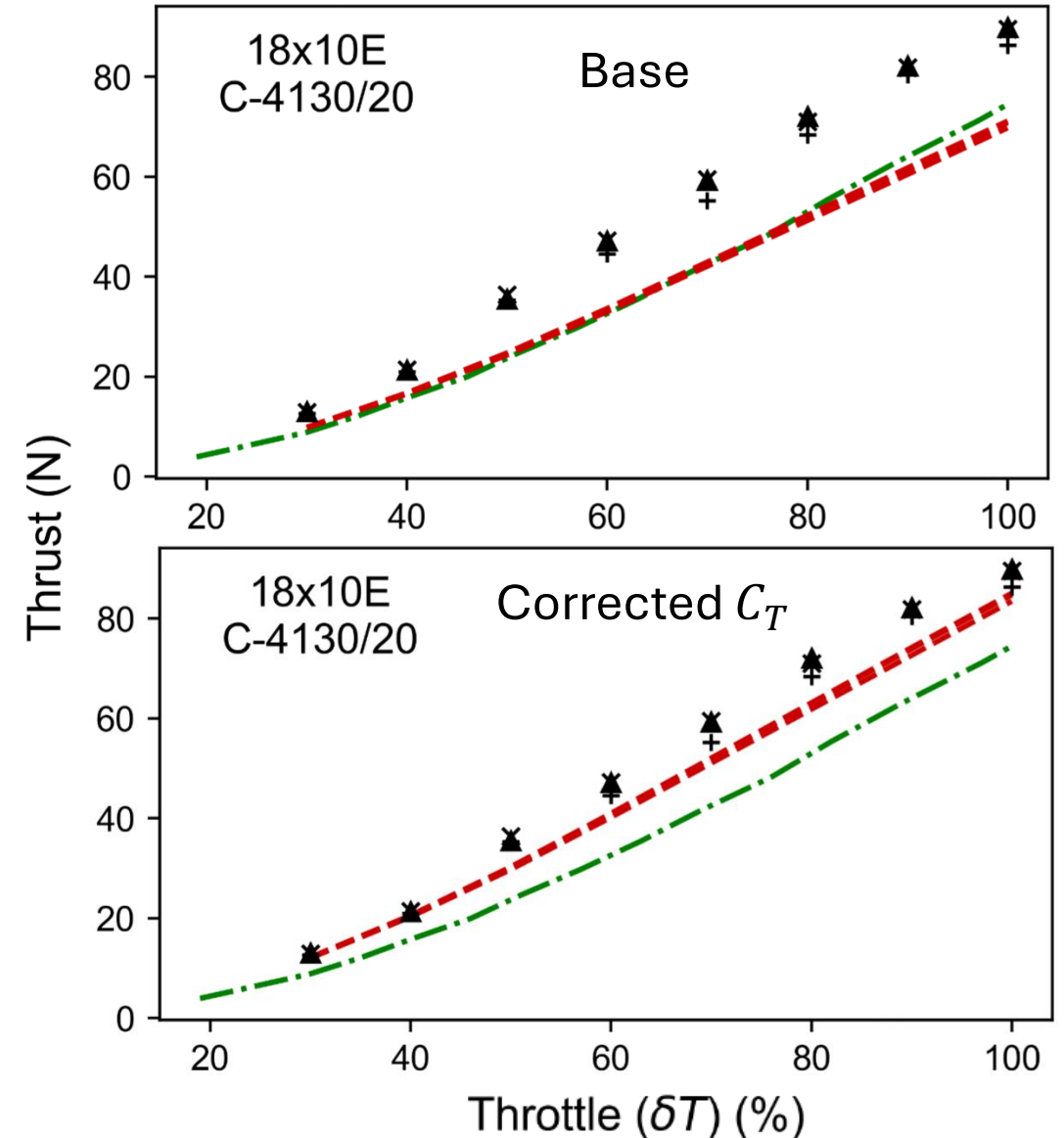
Static Test Results



Static Test Results

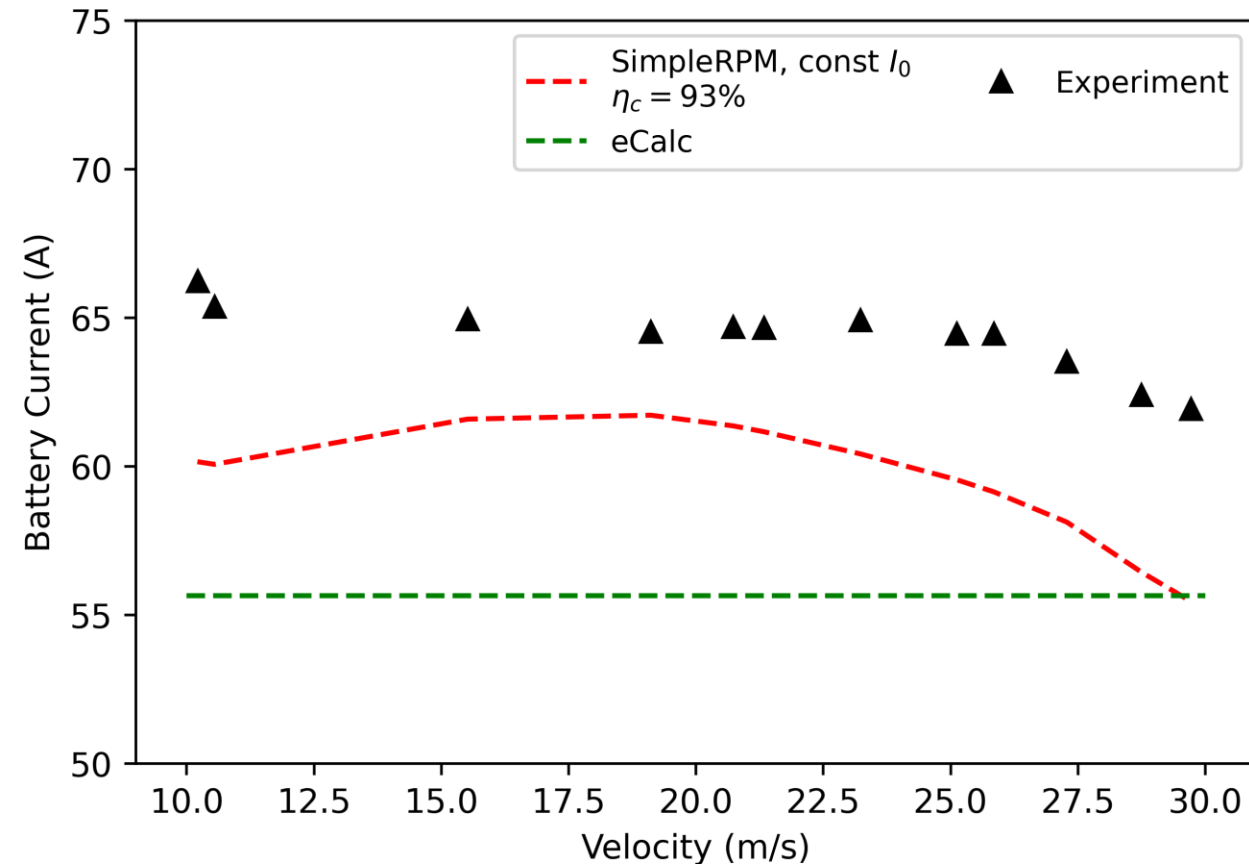
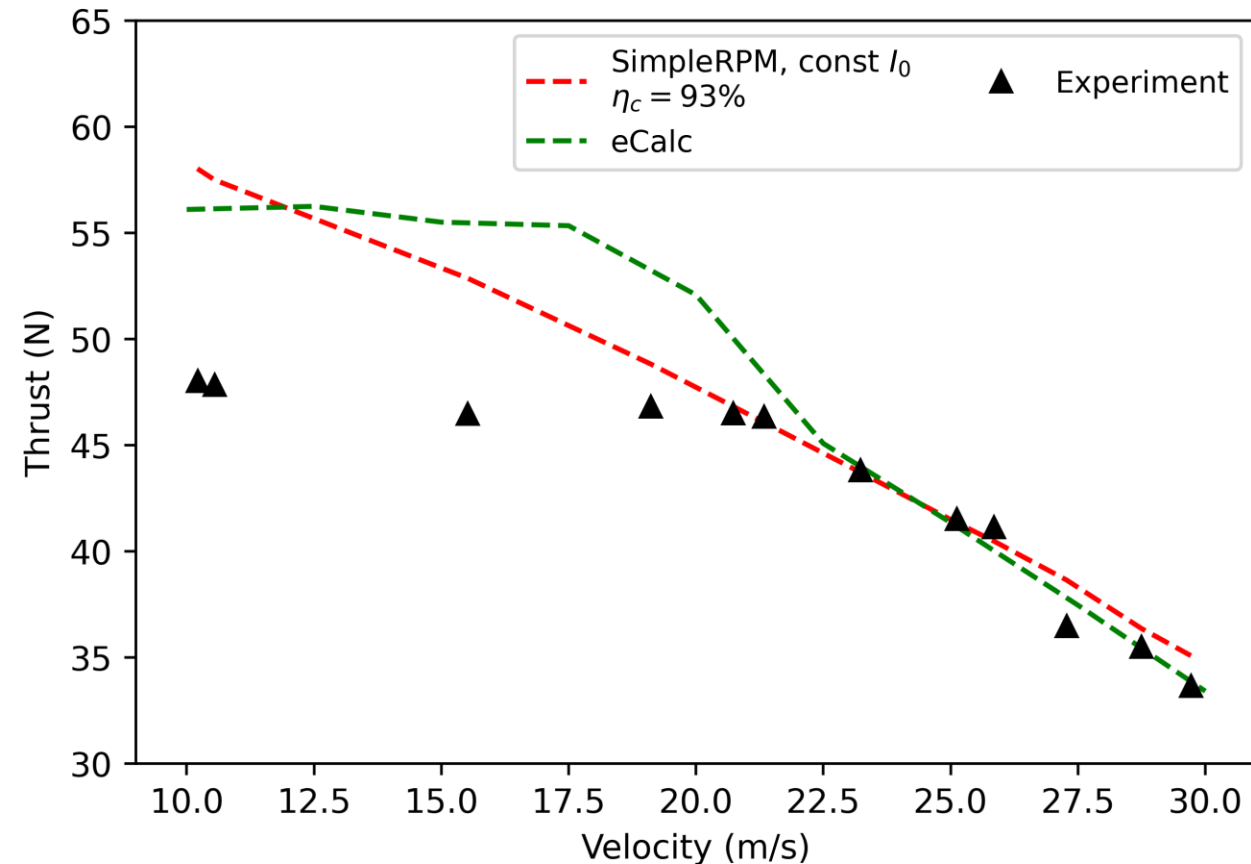
Overall:

- $<15\%$ I_b error
- $<3\%$ RPM error
- $<30\%$ Thrust error base
- $<15\%$ Thrust error with corrected C_T
- I_b , Thrust prediction on par with eCalc at base
- RPM prediction and corrected Thrust significantly improved versus eCalc



Dynamic Test Results

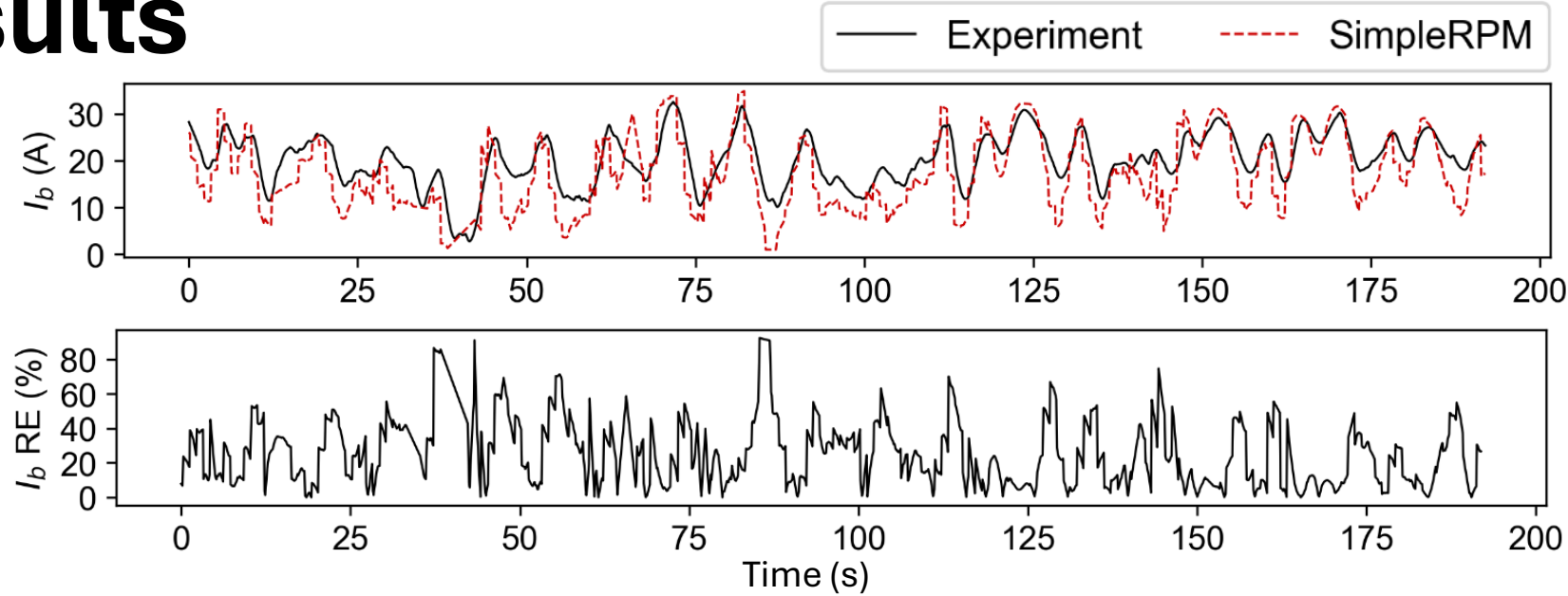
- Accurate thrust prediction by SimpleRPM for unstalled condition
- Significant improvement to I_b prediction ($\sim 7\%$ RE) over eCalc ($\sim 15\%$ RE)



Flight Test Results

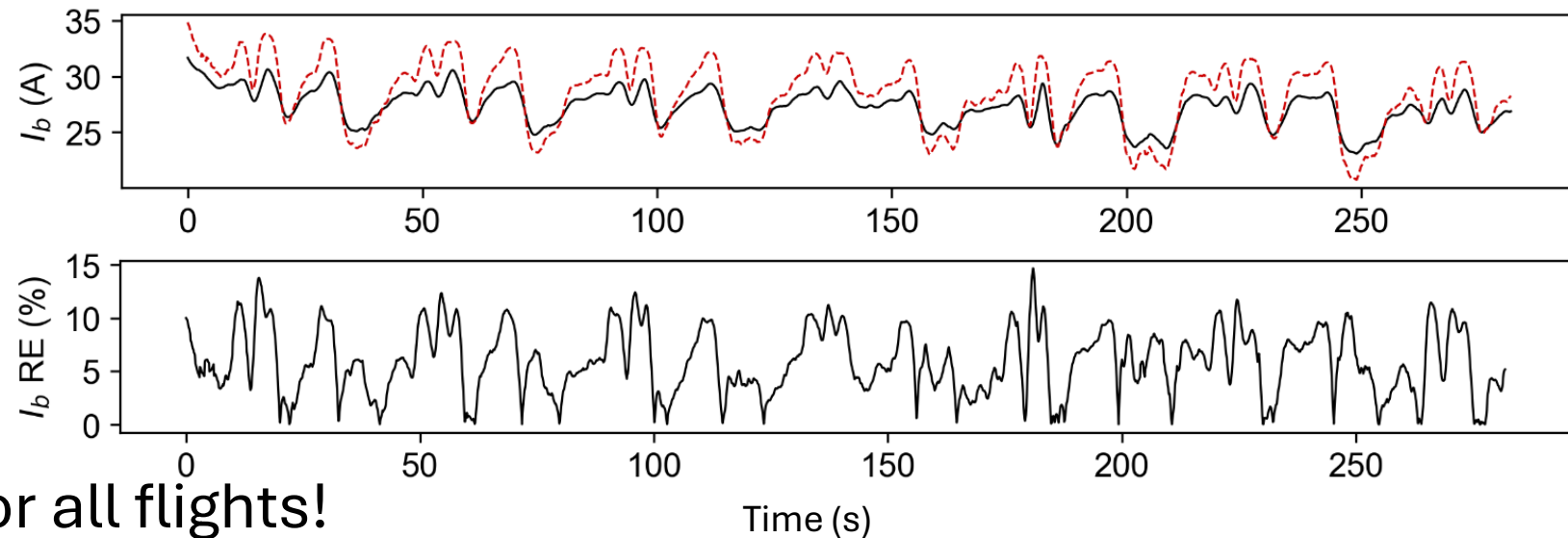
Flight 1:

- Clean config maiden, constant δT adjustments, more dynamic maneuvers
- Max SOC error: 5.5%



Flight 3:

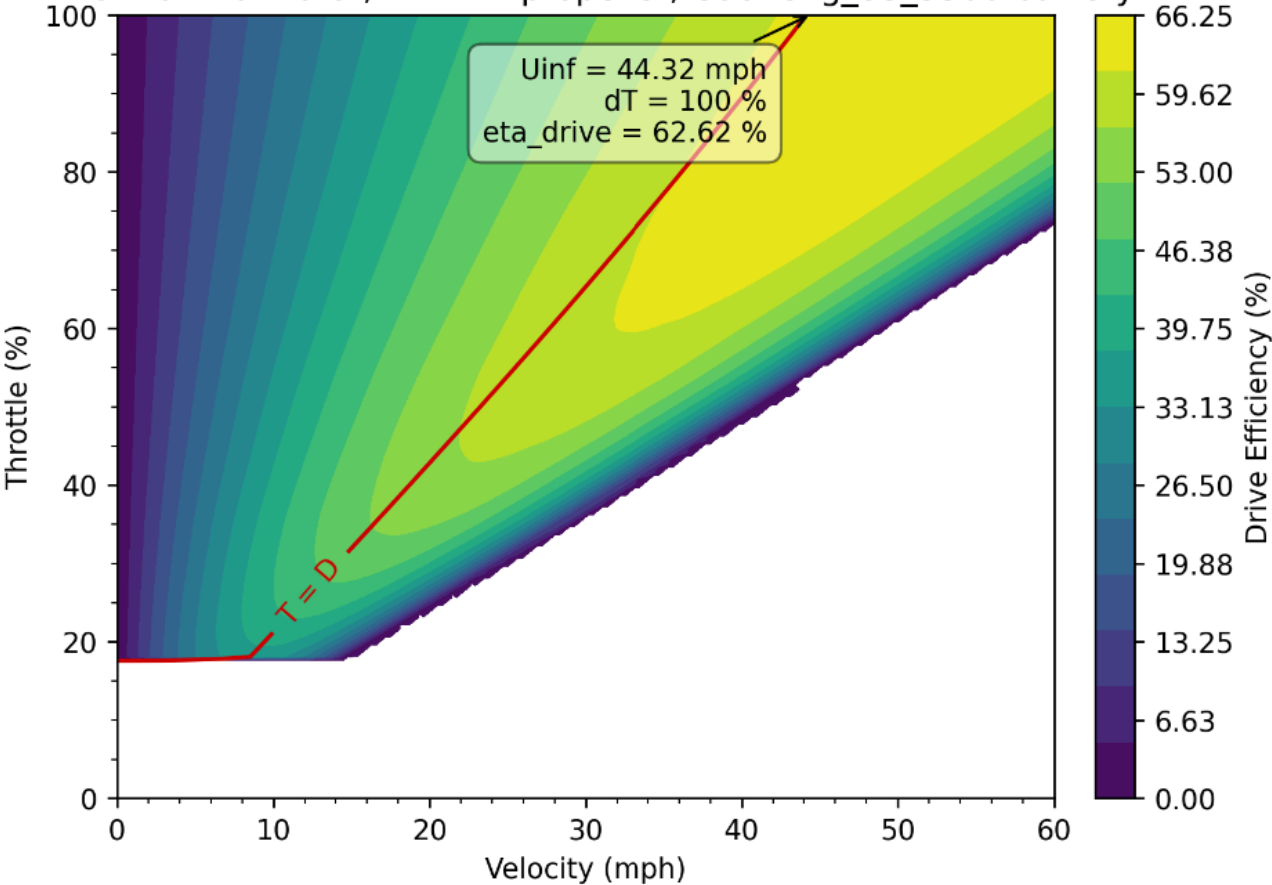
- Banner flight, δT fixed, competition flight course
- Max SOC error: 1.6%



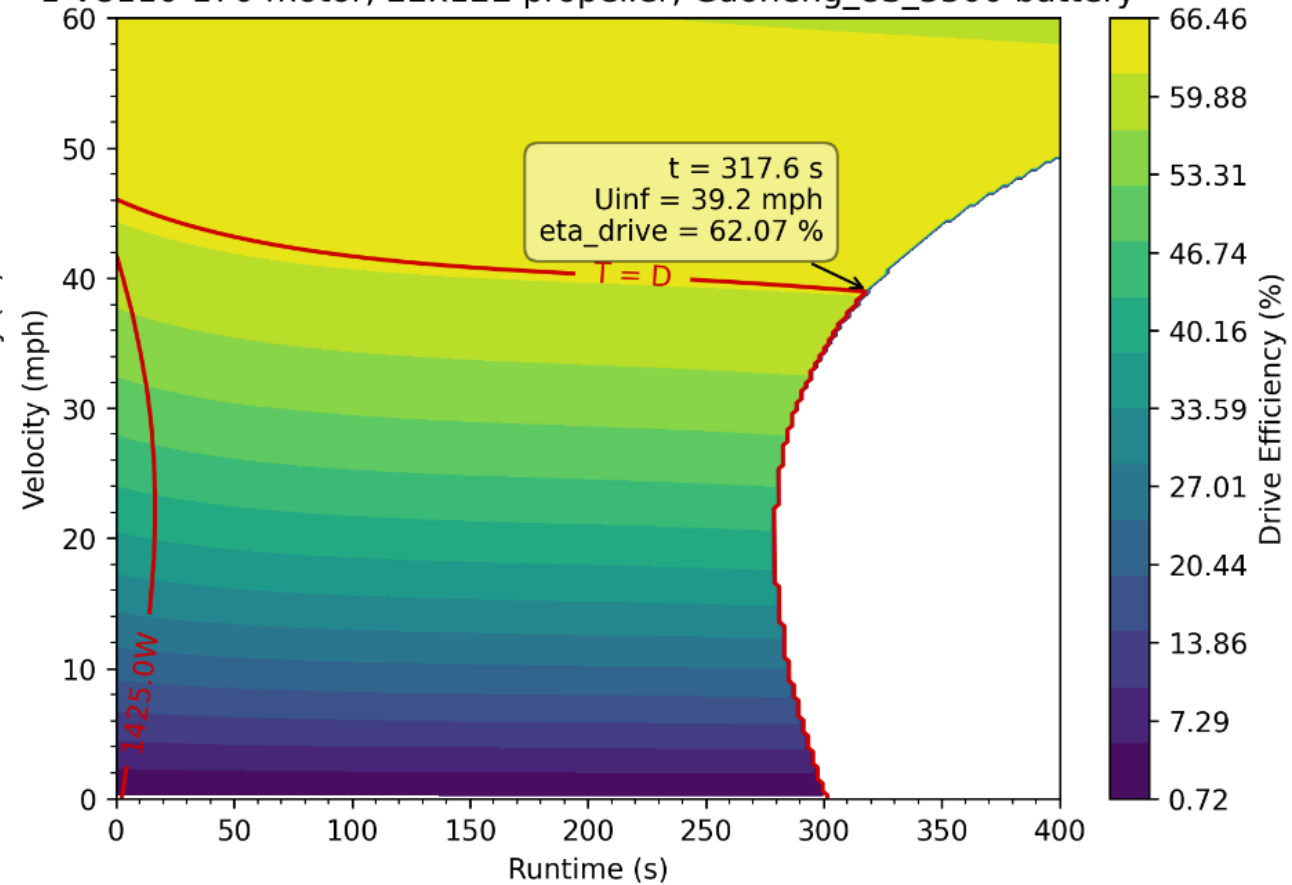
Max RPM error was 3% for all flights!

UAVDEX Design Spaces

Velocity (mph), Throttle (%) sweeps; $t = 30$ s, $\rho = 1.17$ kg/m³
 1 V8110-170 motor, 22x12E propeller, Gaoneng_8S_3300 battery



Runtime (s), Velocity (mph) sweeps; $dT = 100$ %, $\rho = 1.17$ kg/m³
 1 V8110-170 motor, 22x12E propeller, Gaoneng_8S_3300 battery



Summary

- Developed a simplified propulsion model (SimpleRPM) for easy implementation in electric aircraft MDAO codes
- Performed model validation with static thrust, dynamic thrust, and flight testing
- Created an accessible Python library (UAVDEX) to aid student design teams in rigorous propulsion selection

